

ELK Stack Interview Questions

Organized & Deduplicated Reference

1. ELK Overview

Q: What is ELK?

ELK stands for Elasticsearch, Logstash, and Kibana. It is used to collect, process, store, search, and visualize logs.

Q: Why is ELK used?

For centralized logging, monitoring, troubleshooting, security analysis, and performance monitoring.

2. Elasticsearch — Core Concepts

Q: What is Elasticsearch?

A distributed search and analytics engine that stores logs as JSON documents and enables fast searching and aggregation.

Q: What is a Cluster?

A group of one or more Elasticsearch nodes working together and sharing the same cluster name.

Q: What is a Node?

A single Elasticsearch instance that stores data and participates in indexing and search operations.

Q: What is an Index?

A logical collection of related JSON documents, similar to a database in a relational system.

Q: What is a Document?

A JSON object that represents a single record or log entry stored in Elasticsearch.

Q: What is a Field?

A key-value pair inside a document — similar to a column in a relational database.

Q: What is a Shard?

A partition of an index. Elasticsearch splits large indexes into multiple shards to distribute data and improve performance/scalability.

Q: What is a Replica?

A copy of a primary shard that provides high availability, fault tolerance, and can improve search performance.

Q: Why is Elasticsearch so fast?

It uses an inverted index, enabling quick full-text search, plus parallel execution across multiple shards.

Q: How does Elasticsearch store data?

Data is stored as JSON documents inside indices. Each index is split into primary shards distributed across nodes; each shard is stored as Lucene indexes (segments) on disk, with optional replica shards for redundancy.

***Interview tip:** A common flow to describe: Application → Filebeat/Logstash → Elasticsearch Index → Primary Shards → Lucene Segments on Disk → Search through Kibana. This shows you understand both logical and physical storage architecture.*

Q: What is the Elasticsearch REST API?

The HTTP interface that allows clients to create, read, update, delete, and search data in Elasticsearch (default port 9200).

3. Replication & Shard Behavior

Q: What is replication in Elasticsearch?

The process of creating one or more replica shards for each primary shard, providing high availability, fault tolerance, and improved search performance.

Q: Why are replicas used?

- Prevent data loss during node failures
- Keep the cluster available
- Improve search performance by distributing search requests
- Increase reliability

Q: Can writes go directly to a replica?

No. All writes go to the primary shard first; once written successfully, the primary replicates the operation to its replicas.

Q: Can searches use replicas?

Yes — Elasticsearch can run search requests on both primary and replica shards to share the load and improve performance.

Q: What happens if a primary shard fails?

Elasticsearch automatically promotes a replica shard to become the new primary, so the cluster keeps operating.

4. Changing Shard Count — Split & Shrink APIs

Q: Can you change the number of primary shards directly on an existing index?

No. You must create a new index using the Split API, Shrink API, or by reindexing with the desired shard configuration.

Q: What is the Split API?

Creates a new index with a larger number of primary shards, distributing data across more shards.

Q: What is the Shrink API?

Creates a new index with fewer primary shards by combining existing ones, reducing overhead for smaller datasets.

Q: Does Split or Shrink modify the existing index?

No. Both create a new index; the original remains unchanged unless later deleted or aliased over.

(These are considered advanced Elasticsearch administration topics, common in mid-level DevOps/Elasticsearch interviews.)

5. Multi-Node Clusters

Q: Can multiple Elasticsearch nodes run on one machine?

Yes, by assigning different HTTP ports, transport ports, data directories, and node names — useful for dev/testing.

Q: Why don't companies run multiple nodes on one server in production?

If that server fails, all nodes on it fail together, eliminating the benefit of high availability. Production clusters spread nodes across multiple machines.

Q: Why are there typically two+ Elasticsearch nodes in a demo/lab setup?

To form a cluster, demonstrate shard replication, improve availability, and simulate a production environment.

Q: Why is cluster.name required?

Nodes with the same cluster.name discover each other and join the same cluster.

Q: What is node.name?

Uniquely identifies an Elasticsearch node within the cluster.

Q: What does discovery.seed_hosts do?

Lists the addresses of other nodes that Elasticsearch contacts during cluster discovery.

Q: How do you verify nodes joined the cluster?

```
curl http://localhost:9200/_cat/nodes?v
```

Q: How do you verify where primary and replica shards are located?

```
curl http://localhost:9200/_cat/shards?v
```

Shows which node holds each primary (p) and replica (r) shard.

6. Logstash

Q: What is Logstash?

An open-source data processing pipeline that collects data from multiple sources, transforms it using filters, and sends it to destinations like Elasticsearch.

Q: What are the three stages of a Logstash pipeline?

- Input – collects data
- Filter – processes and transforms data
- Output – sends data to Elasticsearch or another destination

Q: Why is Logstash used?

To collect logs from multiple sources, structure them, and send them to Elasticsearch for storage and analysis.

Q: Can Logstash work without Elasticsearch?

Yes — it can send data to many destinations (files, Kafka, Redis, Amazon S3, databases, console). Elasticsearch is just the most common output.

7. Kibana

Q: What is Kibana?

The visualization and analytics tool of the Elastic Stack. It reads data from Elasticsearch and lets users search logs, build dashboards/visualizations, monitor applications, and generate alerts.

Q: Does Kibana store data?

No — it reads and displays data from Elasticsearch; it doesn't store data itself.

Q: What is the Discover feature?

Used to search, filter, and inspect logs stored in Elasticsearch — useful for troubleshooting and log analysis.

Q: What are Kibana dashboards?

Combine multiple visualizations, charts, and metrics into a single screen for real-time monitoring.

Q: What is Kibana Dev Tools?

A built-in console for running Elasticsearch REST API commands directly, without curl or Postman.

Q: Can Kibana generate alerts?

Yes, based on predefined conditions (e.g., high CPU, increased error rates, service failures), notifying via channels like email or Slack.

8. Beats & Filebeat

Q: What are Beats?

Lightweight agents that collect logs, metrics, or other data from servers and send them to Logstash or Elasticsearch.

Q: What is Filebeat?

A lightweight log shipper that reads log files and forwards them to Logstash or Elasticsearch. It's the Beat used for application logs.

Q: What is Metricbeat used for?

Collecting system/application performance metrics — CPU, memory, disk, network usage.

Q: Why use Filebeat instead of running Logstash on every server?

Filebeat is lightweight and resource-efficient. It ships data from many servers to a centralized Logstash instance (or directly to Elasticsearch), reducing resource usage and simplifying management.

9. X-Pack

Q: What is X-Pack?

The built-in feature set of the Elastic Stack, providing security, monitoring, alerting, reporting, machine learning, and other advanced capabilities.

Q: Is X-Pack a separate plugin?

In older Elastic Stack versions, yes. In modern/current versions, its features are integrated by default and can be enabled/configured as needed.

10. Ports Reference

Port	Service
9200	Elasticsearch REST API (HTTP, used by clients/Kibana/apps)
9300	Elasticsearch transport protocol (internal node-to-node cluster communication)
5601	Kibana
5044	Logstash Beats input (e.g., from Filebeat)
5000	Logstash TCP/UDP input

Q: Which port does Filebeat listen on?

None — Filebeat doesn't listen on a port. It sends data to Logstash (typically port 5044) or directly to Elasticsearch (9200).

Q: Which port(s) does Logstash use?

Depends on the configured input plugin — common ones are 5044 (Beats), 5000 (TCP/UDP), and other configurable ports (e.g., 8080 for HTTP input).

11. Docker Compose for ELK

Q: What is Docker Compose?

A tool for defining and running multi-container Docker applications using a YAML file.

Q: Why do we use version: "3.8"?

To specify the Docker Compose file format version and its supported features.

Q: Why is `discovery.type=single-node` required?

It tells Elasticsearch to run as a standalone node instead of expecting a full cluster.

Q: Why disable `xpack.security` in development?

To simplify setup by removing authentication and TLS. In production, security should be enabled.

Q: What is `ES_JAVA_OPTS`?

Configures the JVM heap size used by Elasticsearch — proper sizing matters for performance and stability.

Q: Why use `depends_on`?

To control container startup order. Note: it does not guarantee that a dependency is fully ready/initialized.

Q: Why use `elasticsearch:9200` instead of `localhost:9200` inside Docker?

Containers communicate over Docker's internal network using service names as hostnames.

Q: Why mount volumes?

To share files between host and container and to persist data (e.g., Elasticsearch data survives container stop/recreate) without rebuilding the image.

Q: Why does Kibana use `ELASTICSEARCH_HOSTS`?

It tells Kibana which Elasticsearch node to connect to for querying data and building visualizations.

Q: What is the difference between an image and a container?

An image is a read-only template containing the application and its dependencies. A container is a running instance created from that image.